

Design and Analysis of Randomized Trials in Public Health

Homework 5

1. Please use the “hw4” SAS dataset on Brightspace to answer the questions below. Before importing the dataset into SAS, run the code below. This will create value labels for your dataset.

```
proc format;  
  value yes12no      1="yes"  
                    2="no";  
  value sex          1="female"  
                    2="male";  
  value trt          1="treatment"  
                    0="control";  
run;
```

These data are from a randomized trial in which eligible participants were randomized to receive a physical activity intervention (“treatment”) or no intervention (“control”). Their body mass index (BMI) was measured at the beginning (“baselineBMI”) and end (“endlineBMI”) of the study. Participants were eligible for the study if they (1) were aged 18-50 years, (2) were evaluated to have a sedentary lifestyle, and (3) provided informed consent. Participants were excluded if they failed to meet any of these criteria.

Does the physical activity intervention affect body mass index? Use “endlineBMI” as your outcome, and this time, analyze your data using the regression approach.

- a) What is your statistical model? Make sure to specify the distribution of your error term and to define all notation you use. (3 points)
 - b) What is your estimated model? Use SAS. (2 points)
 - c) Explain in words how you interpret the intercept of your estimated model. (2 points)
 - d) Explain in words how you interpret the slope of your estimated model. (2 points)
2. You conduct a study with a completely randomized design and two arms, an intervention group and a control group. You analyze your data using an ANOVA approach.
 - a) Here is the ANOVA table you generate, with some numbers missing. Fill in the missing cells (some of which will remain blank). (3 points)

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model		1435.30			0.0409
Error					
Corrected Total	158	54458.92			

- b) In total, how many people participated in your study? (1 point)
- c) When using your F statistic to calculate the p-value, what distribution do you use? (Don't forget to specify parameters, e.g., degrees of freedom, if applicable.) (2 points)

Homework 5

Alexandra Chang

PUBH 526: Design and Analysis of Randomized Trials in Public Health

October 9, 2025

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Does the physical activity intervention affect body mass index? Use “endlineBMI” as your outcome, and this time, analyze your data using the regression approach.

Question 1.a (3 points)

What is your statistical model? Make sure to specify the distribution of your error term and to define all notation you use.

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim \text{iid } N(0, \sigma^2)$$

- Y_i = endline BMI for participant i ,
- $X_i = 1$ if the participant i received the physical activity intervention (treatment), and 0 if in the control group,
- β_0 = mean endline BMI for the control group,
- β_1 = mean difference in endline BMI between treatment and control groups.

Question 1.b (2 points)

What is your estimated model? Use SAS.

- $\text{Treatment}_i \in \{0, 1\}$ (1 = treatment, 0 = control), the fitted model is

$$\widehat{\text{endlineBMI}}_i = 26.024 + 0.606 \text{treatment}_i.$$

(SAS PROC GLM: $n = 403$, $F(1, 401) = 9.61$, $p = 0.0021$, $R^2 = 0.0234$, Root MSE = 1.955.)

Question 1.c (2 points)

Explain in words how you interpret the intercept of your estimated model.

- The intercept, $\hat{\beta}_0 = 26.02$, represents the estimated **mean endline BMI for participants in the control group**, which is the baseline value.
 - Baseline is the predicted BMI when treatment = 0, meaning no physical activity intervention was received.

Question 1.d (2 points)

Explain in words how you interpret the slope of your estimated model.

- The slope, $\hat{\beta}_1 = 0.61$, represents the **estimated mean difference** in endline BMI between the treatment (1) and control (0) groups.
- According to the SAS results, the slope estimate is positive and statistically significant ($p = 0.0021$), suggesting that on average participants in the **treatment group had a 0.61-unit higher BMI** at endline compared to the control group.
 - Thus, the **intervention did not reduce BMI** as expected.

You conduct a study with a completely randomized design and two arms, an intervention group and a control group. You analyze your data using an ANOVA approach.

Question 2.a (3 points)

Here is the ANOVA table you generate, with some numbers missing. Fill in the missing cells (some of which will remain blank).

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	?	1435.30	?	?	0.0409
Error	?	?	?		
Total	158	54458.92			

Step-by-step calculations:

- **DF for Model:** This is a simple comparison between two groups (treatment vs. control), so there is only one predictor.

$$DF_{\text{Model}} = 1$$

- **DF for Error:**

$$DF_{\text{Error}} = DF_{\text{Total}} - DF_{\text{Model}} = 158 - 1 = 157$$

- **Sum of Squares for Error:**

$$SS_{\text{Error}} = SS_{\text{Total}} - SS_{\text{Model}} = 54458.92 - 1435.30 = 53023.62$$

- **Mean Square for Model:**

$$MS_{\text{Model}} = \frac{SS_{\text{Model}}}{DF_{\text{Model}}} = \frac{1435.30}{1} = 1435.30$$

- **Mean Square for Error:**

$$MS_{\text{Error}} = \frac{SS_{\text{Error}}}{DF_{\text{Error}}} = \frac{53023.62}{157} = 337.68$$

- **F Statistic:**

$$F = \frac{MS_{\text{Model}}}{MS_{\text{Error}}} = \frac{1435.30}{337.68} = 4.25$$

Completed ANOVA Table:

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1435.30	1435.30	4.25	0.0409
Error	157	53023.62	337.68		
Total	158	54458.92			

Question 2.b (1 point)

In total, how many people participated in your study?

- The total degrees of freedom (DF) in the ANOVA table is 158. Since the total DF is defined as:

$$\begin{aligned} \text{DF}_{\text{Total}} &= n - 1 \\ n &= \text{DF}_{\text{Total}} + 1 = 158 + 1 = \boxed{159} \end{aligned}$$

- Therefore, the study had **159 participants** in total.

Question 2.c (2 points)

When using your F statistic to calculate the p-value, what distribution do you use? (Don't forget to specify parameters, e.g., degrees of freedom, if applicable.)

- The F statistic used to test for a difference between group means follows an F distribution under the null hypothesis.
- The two parameters of the F distribution correspond to the degrees of freedom for the numerator (Model) and the denominator (Error). From the ANOVA table:

$$F \sim F(1, 157)$$

- Hence, the test statistic follows an **F distribution with 1 and 157 degrees of freedom**.

See SAS codes/outputs below:

```
1  /*****
2  * PUBH 526      Design and Analysis of Randomized Trials in Public
3  * Homework 5
4  * Alexandra Chang
5  * Dataset: hw4.sas7bdat
6  *****/
7
8  libname mydata '/home/u64258127/sasuser.v94';
9
10 proc format;
11     value yes12no 1="yes" 2="no";
12     value sex 1="female" 2="male";
13     value trt 1="treatment" 0="control";
14 run;
15
16 /*****
17 * Question 2: Fit simple linear regression model
18 *****/
19 proc glm data=mydata.hw4;
20     class treatment;
21     model endlineBMI = treatment / solution;
22     format treatment trt.;
23     title "Linear Regression Model for Endline BMI by Treatment
24           Group";
25 run;
quit;
```

Listing 1: SAS Code for Homework 5

Linear Regression Model for Endline BMI by Treatment Group

The GLM Procedure

Class Level Information		
Class	Levels	Values
treatment	2	control treatment

Number of Observations Read	500
Number of Observations Used	403

Linear Regression Model for Endline BMI by Treatment Group

The GLM Procedure

Dependent Variable: endlineBMI Endline BMI

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	36.732379	36.732379	9.61	0.0021
Error	401	1532.262757	3.821104		
Corrected Total	402	1568.995136			

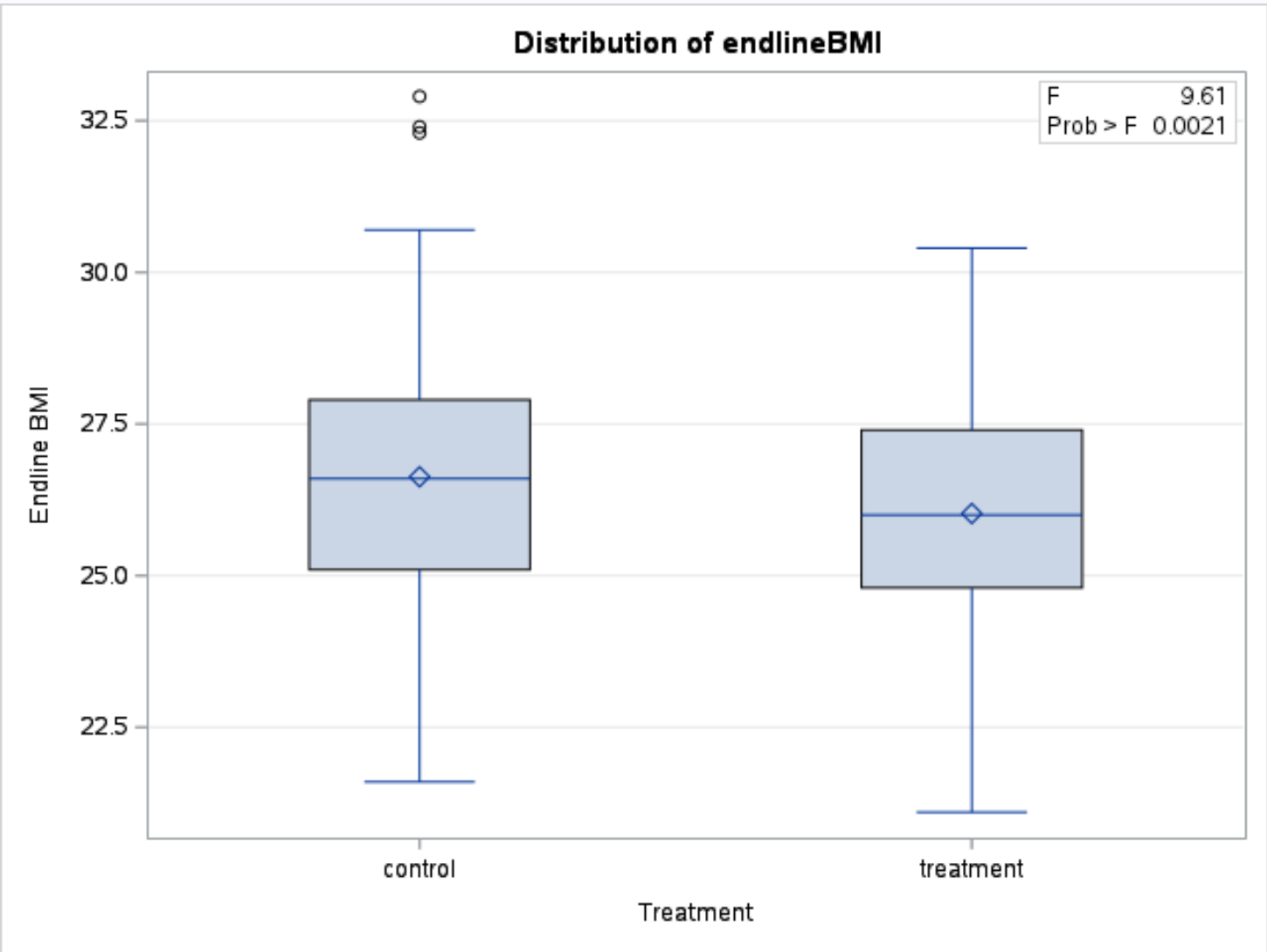
R-Square	Coeff Var	Root MSE	endlineBMI Mean
0.023411	7.431583	1.954764	26.30347

Source	DF	Type I SS	Mean Square	F Value	Pr > F
treatment	1	36.73237918	36.73237918	9.61	0.0021

Source	DF	Type III SS	Mean Square	F Value	Pr > F
treatment	1	36.73237918	36.73237918	9.61	0.0021

Parameter	Estimate		Standard Error	t Value	Pr > t
Intercept	26.02396313	B	0.13269806	196.11	<.0001
treatment control	0.60560676	B	0.19532625	3.10	0.0021
treatment treatment	0.00000000	B	.	.	.

Note: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.



Design and Analysis of Randomized Trials in Public Health

Homework 5 Solutions

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Does the physical activity intervention affect body mass index? Use “endlineBMI” as your outcome, and this time, analyze your data using the regression approach.

- a) What is your statistical model? Make sure to specify the distribution of your error term and to define all notation you use. (3 points)

$$Y_j = \beta_0 + \beta_1 X_j + \varepsilon_j$$
$$\varepsilon_j \sim iid N(0, \sigma^2)$$

$j = 1, 2, \dots, 403$ (an index for the 403 people who were enrolled and for whom you have data)

$X_j =$ 1 if the j th observation is in the intervention group
0 otherwise

Y_j is the endline BMI for the j th observation

- b) What is your estimated model? Use SAS. (2 points)

Because of the way I defined X_i above (making the control treatment my reference), I'll estimate my model this way:

```
proc glm data=hw4 plots=all;
  class treatment(ref="control");
  model endlineBMI = treatment / solution;
run;
```

This gives me the following estimated model:

$$\hat{Y} = 26.63 - 0.61X$$

- c) Explain in words how you interpret the intercept of your estimated model. (2 points)

This is the average BMI I expect for people in this population who do not receive the intervention.

- d) Explain in words how you interpret the slope of your estimated model. (2 points)

This is the change in BMI I expect for people in this population who receive the intervention, compared with people who do not receive it. This slope is statistically significant, meaning that with intervention, I expect people's BMI to be 0.61 units lower on average if they receive this intervention, compared with the average of people who don't receive the intervention.

2. You conduct a study with a completely randomized design and two arms, an intervention group and a control group. You analyze your data using an ANOVA approach.

- a) Here is the ANOVA table you generate, with some numbers missing. Fill in the missing cells (some of which will remain blank). (3 points)

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1435.29611	1435.29611	4.25	0.0409
Error	157	53023.62213	337.73008		
Corrected Total	158	54458.91824			

- b) In total, how many people participated in your study? (1 point)

159

- c) When using your F statistic to calculate the p-value, what distribution do you use? (Don't forget to specify parameters, e.g., degrees of freedom, if applicable.) (2 points)

You use an $F_{1,157}$ distribution (an F distribution with numerator df=1 and denominator df=157).